

# Sauhardha Kafle

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## Skills

**Languages:** Rust, Python, JavaScript, TypeScript

**Backend:** Actix-web, NodeJS, ExpressJS, AIOHTTP, Flask, FastAPI

**Frontend:** NextJS, ReactJS, TailwindCSS,

**Databases & Tools:** Redis, MySQL, PostgreSQL, MongoDB, Appwrite

**AI/ML:** Deep Learning, Natural Language Processing, RAG

**Infra:** Docker, gRPC, IPC, webRTC, websocket

## Education

**Kathmandu University**

*B.E. in Computer Engineering*

**Dec 2022 – Feb 2027**

*Dhulikhel, Nepal*

**Orchid Science College**

*Higher Secondary (+2), Science*

**2020 – 2022**

*Chitwan, Nepal*

**Vishwa Jyoti Secondary School**

*School Leaving Certificate (SLC)*

**2019**

*Nawalpur, Nepal*

## Experience

**AI/LLM Fellowship | KonnectCraft**

**2026**

- Built the ingestion pipeline for a hybrid RAG platform: parsed unstructured documents, extracted entities and relations into a **Neo4j** knowledge graph, and indexed embeddings for combined graph and vector retrieval.
- Designed an embeddable, **multi-tenant** chatbot system that lets clients add custom RAG-powered assistants to their own applications, with per-client data isolation.

## Open Source Contributions

[Dora-rs](#) | *Rust*

- PR #1341: Implemented global environment variables in dataflow YAML, allowing variables to be inherited by all nodes by default while also preserving node-level overrides.
- PR #1609: Fixed a pointer boundary check that was rejecting buffers starting at the beginning of a memory region.
- Issue #56 – Identified and fixed an incorrect Conda environment name in the Dora documentation.

[St. Jude Rust Labs](#) | *Rust*

- PR #9 – Replaced the deprecated **async-std** runtime with **smol**, updating integration, feature flags, and CI configuration.

## Projects

[Torrex](#) | *Rust* | [Demo](#)

- Built a BitTorrent client protocol from scratch in **Rust**, and a REST API based web dashboard for managing downloads.
- Supports magnet links, torrent files and URLs, with concurrent peer connections, piece verification, pause/resume, and real-time download and peer statistics.
- Implemented Bencode parsing, magnet metadata exchange and HTTP tracker discovery using **Actix Web** and **WebSockets**.

[nutshell](#) | *Rust, Python, TypeScript, Next.js* | [Demo](#)

- Built a Vercel-like platform that takes a GitHub repository, builds it and deploys it automatically.
- Designed the deployment pipeline as separate services, completing the clone, build and deployment process in ~35 seconds.
- Used **Cloudflare R2** for source storage, **Redis** for job queues, isolated **Docker** containers for builds, and **Rust/Actix Web** to serve deployed applications.

[Bird Sound Classifier](#) | *Python, PyTorch*

- Built a deep learning pipeline to identify bird species from audio recordings, focused on birds found in Nepal and the Himalayas.

- Automated the dataset pipeline by collecting recordings from Xeno-Canto, filtering low-quality samples, and preparing train, validation and test datasets.
- Fine-tuned **AST** (Audio Spectrogram Transformer) on log-Mel spectrograms with augmentation and class-balanced loss to handle differences in species and recording quality.

[Aether](#) | *Python, Rust, WebRTC* | [Demo](#)

- Built a browser-based peer-to-peer remote desktop and compute-sharing platform.
- Streams a host's screen and sends remote mouse and keyboard input in real time using **WebRTC**.
- Built the signaling layer with **FastAPI**, Redis Pub/Sub and WebSockets, with a Next.js frontend, PostgreSQL and Docker Compose.

[CodeRoom](#) | *Node.js, React, MySQL, Docker* | [Demo](#)

- Built a real-time collaborative web IDE, supporting multi-user editing, chat and code execution in JavaScript, Python and C++.
- Designed and built an isolated **Docker**-based execution system with ~0.2s compile latency.
- Used **Express**, Socket.io, JWT authentication, OTP email verification and MySQL for the backend, with Monaco Editor on the frontend.

[dlm\\_rust](#) | *Rust* | [Demo](#)

- Built an asynchronous terminal download manager in **Rust** with a separate client and background daemon.
- Supports multiple concurrent downloads with pause, resume and automatic retry, controlled through an interactive terminal interface.
- Used **Unix Domain Sockets** for communication between the client and daemon, with **Tokio** for asynchronous tasks and **Ratatui** for the terminal UI.

[Kartru](#) | *Rust, Tokio, LLM APIs*

- Built a CLI coding agent in **Rust** that plans and completes development tasks autonomously through an LLM-driven tool-use loop.
- Implemented file-read, file-write and shell-execution tools that the model invokes across multiple steps until the task is complete.
- Used **Tokio** for async execution and integrated models through the **OpenRouter** API.

## Leadership & Activities

**Technical Lead** | *ITMeet* 2025

- Led a 10-member technical team in developing and deploying the official ITMeet 2025 website using React and Vite, coordinating development and delivery of participant-facing event features.
- Our team involved in development of interactive AR experiences, including a MindAR/Three.js image-tracking marketing experience and a GPS-based AR venue navigation system with 3D directional guidance, distance tracking, and arrival detection.
- Managed event-wide technical infrastructure and on-site operations, ensuring reliable internet connectivity and coordinating technical support throughout the event.

**Event Manager** | *IdeaPitching* 2024

Managed end-to-end event operations, including judge, participants coordination, venue planning, scheduling, and on-site execution.

**Technical Member** | *ITMeet* 2024

Contributed to the development of the ITMeet event website and participant-facing functionality.

**Volunteer** | *ITMeet* 2023

**Volunteer** | *Aavishkar — KURC* 2023

## Achievements

**Winner** | *Ecothon* 2026

- Built a forest health monitoring system that uses bird sounds to estimate the health of a forest ecosystem from audio recordings.
- Used **BirdNET 2.4** for bird species detection and combined biodiversity, rarity, native-species and forest-dependency data from GBIF, IUCN and EltonTraits to calculate a Forest Health Quality Index; added a **RAG** pipeline with **FAISS**, sentence-transformers and LLM to provide additional ecological context for detected species.
- Won the **Open Innovation Track** at Ecothon 2026 with the project, competing against other teams developing solutions around environmental and sustainability challenges.